

```

* =====.
* BI and Decision-Making Quality in SMEs - SPSS analysis (n = 178).
* Source: Google Form export of 188 valid responses; 1 removed for
* contradictory consent and 9 for careless responding (identical answer
* to all 17 items) -> n = 178. No missing data; all items in range.
* Output is the document of record. Validity analyses SPSS cannot do
* (CFA, HTMT, AVE/CR, Fornell-Larcker) are in BI_DMQ_validity.py
* =====.
* Coding: Company_Size 1=<50 2=50-99 3=99-250(medium).
*           Job_Position 1=CEO 2=Senior 3=Middle 4=Operational 5=Other.
*           Industry 1=Manufacturing 2=Retail 3=Service 4=Technology 5=Other.
*           Items: 1=Strongly Disagree ... 5=Strongly Agree.
* =====.

```

```

* =====.
* LOAD DATA. Easiest: open BI_DMQ_data_coded.sav in SPSS first, then
* run this file from the COMPUTE lines below. Or edit the path here to
* the full location of the .sav and run the whole file.
* =====.

```

```
GET FILE='BI_DMQ_data_coded.sav'.
```

```

>Error # 61 in column 10. Text: BI_DMQ_data_coded.sav
>The filename is not valid.
>Execution of this command stops.
DATASET NAME DataSet178 WINDOW=FRONT.

```

```

COMPUTE Strategic_Mean = MEAN(Strategic_BI_Availability,Strategic_BI_Alternatives,Strategic_BI_Alignment).
COMPUTE Tactical_Mean = MEAN(Tactical_BI_Availability,Tactical_BI_Resource_Alloc,Tactical_BI_Data_Decision).
COMPUTE Operational_Mean= MEAN(Operational_BI_Availability,Operational_BI_Agility,Operational_BI_Quality).
COMPUTE Decision_Making_Quality= MEAN(BI_Improves_Decisions,BI_Speed_Accuracy,BI_Data_Inform,BI_Alternative_Solution).
EXECUTE.

```

```

* ===== STEP A0: data screening (range + univariate outliers). =====.
* (protocol items 3-4: z-scores beyond +/-3.29 flag univariate outliers).
DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality

```

/SAVE /STATISTICS=MEAN STDDEV MIN MAX.

Descriptives

Notes		
Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality /SAVE /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00
Variables Created or Modified	ZStrategic_Mean	Zscore(Strategic_Mean)
	ZTactical_Mean	Zscore(Tactical_Mean)
	ZOperational_Mean	Zscore (Operational_Mean)
	ZDecision_Making_Quality	Zscore (Decision_Making_Quality)

[DataSet178] C:\Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Strategic_Mean	178	1.00	5.00	3.7341	.77820
Tactical_Mean	178	1.00	5.00	3.7978	.70798
Operational_Mean	178	1.33	5.00	3.8258	.74679
Decision_Making_Quality	178	1.50	5.00	3.8244	.75334
Valid N (listwise)	178				

* ===== STEP 0: demographics incl. firm size (Table 1). =====.
 FREQUENCIES VARIABLES=Company_Size Job_Position Industry.

Frequencies

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \\Users\\Hoda\\OneDrive\\De sktop\\GIT\\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Company_Si ze Job_Position Industry.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Statistics

		Company_Size	Job_Position	Industry
N	Valid	178	178	178
	Missing	0	0	0

Frequency Table

Company_Size

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	65	36.5	36.5	36.5
	2	62	34.8	34.8	71.3
	3	51	28.7	28.7	100.0
	Total	178	100.0	100.0	

Job_Position

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	13	7.3	7.3	7.3
	2	21	11.8	11.8	19.1
	3	36	20.2	20.2	39.3
	4	30	16.9	16.9	56.2
	5	78	43.8	43.8	100.0
	Total	178	100.0	100.0	

Industry

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	13	7.3	7.3	7.3
	2	46	25.8	25.8	33.1
	3	52	29.2	29.2	62.4
	4	30	16.9	16.9	79.2
	5	37	20.8	20.8	100.0
	Total	178	100.0	100.0	

```
* ===== STEP 1: reliability+ item-total+ alpha-if-deleted =====.
RELIABILITY VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStra
tegic_BI_Alignment
/SCALE('Strategic BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
```

Reliability

Notes		
Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment /SCALE('Strategic BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.01

Scale: Strategic BI

Case Processing Summary

		N	%
Cases	Valid	178	100.0
	Excluded ^a	0	.0
	Total	178	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.795	3

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Strategic_BI_Availability	7.56	2.610	.613	.749
Strategic_BI_Alternatives	7.46	2.532	.698	.654
Strategic_BI_Alignment	7.39	2.871	.606	.753

```
RELIABILITY VARIABLES=Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_BI_Data_Decisions
/SCALE('Tactical BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisions /SCALE('Tactical BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Scale: Tactical BI

Case Processing Summary

		N	%
Cases	Valid	178	100.0
	Excluded ^a	0	.0
	Total	178	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.760	3

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Tactical_BI_Availability	7.69	2.183	.557	.720
Tactical_BI_Resource_Alloc	7.62	2.225	.637	.626
Tactical_BI_Data_Decisions	7.48	2.330	.581	.689

RELIABILITY VARIABLES=Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality
/SCALE('Operational BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.

Reliability

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=Operational_ BI_Availability Operational_BI_Agility Operational_BI_Quality /SCALE('Operational BI') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.01

Scale: Operational BI

Case Processing Summary

		N	%
Cases	Valid	178	100.0
	Excluded ^a	0	.0
	Total	178	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.773	3

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Operational_BI_Availability	7.69	2.341	.590	.720
Operational_BI_Agility	7.62	2.552	.632	.671
Operational_BI_Quality	7.65	2.558	.608	.694

```
RELIABILITY VARIABLES=BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformB  
I_Alternative_Solutions  
/SCALE('DMQ') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /SCALE('DMQ') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Scale: DMQ

Case Processing Summary

		N	%
Cases	Valid	178	100.0
	Excluded ^a	0	.0
	Total	178	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.843	4

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
BI_Improves_Decisions	11.44	5.570	.670	.804
BI_Speed_Accuracy	11.52	4.918	.735	.774
BI_Data_Inform	11.35	5.460	.679	.800
BI_Alternative_Solutions	11.58	5.555	.629	.821

```
RELIABILITY VARIABLES=BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue  
/SCALE('Challenges') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /SCALE('Challenges') ALL /MODEL=ALPHA /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Scale: Challenges

Case Processing Summary

		N	%
Cases	Valid	178	100.0
	Excluded ^a	0	.0
	Total	178	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.635	4

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
BI_Training_Limit	10.31	4.689	.426	.557
BI_Cost_Barrier	10.55	4.712	.345	.619
BI_Data_Integration_Issue	10.60	4.796	.451	.543
BI_Flexibility_Issue	10.75	4.529	.445	.543

* ===== STEP 2: factorability+ EFA + SAVE factor scores. =====.

* KMO, Bartlett, communalities loadings, % variance; saves Strategic1 etc.

FACTOR VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStrategic_BI_Alignment

/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL Strategic).

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL Strategic).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	2028 (1.980K) bytes
Variables Created	Strategic1	Regression factor score 1

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.691
Bartlett's Test of Sphericity	Approx. Chi-Square	164.646
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Strategic_BI_Availability	.703
Strategic_BI_Alternatives	.860
Strategic_BI_Alignment	.696

Extraction Method: Principal Axis
Factoring.

a. 1 factors extracted. 14 iterations required.

Communalities

	Extraction
Strategic_BI_Availability	.494
Strategic_BI_Alternatives	.740
Strategic_BI_Alignment	.485

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.718	57.267	57.267

Extraction Method: Principal Axis Factoring.

```
FACTOR VARIABLES=Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_
BI_Data_Decisions
/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTAT
E /SAVE REG(ALL Tactical).
```

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:36
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisions /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL Tactical).
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.03
	Maximum Memory Required	2028 (1.980K) bytes
Variables Created	Tactical1	Regression factor score 1

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.686
Bartlett's Test of Sphericity	Approx. Chi-Square	133.094
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Tactical_BI_Availability	.655
Tactical_BI_Resource_Alloc	.804
Tactical_BI_Data_Decisions	.699

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 12 iterations required.

Communalities

	Extraction
Tactical_BI_Availability	.429
Tactical_BI_Resource_Alloc	.646
Tactical_BI_Data_Decisions	.489

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.564	52.144	52.144

Extraction Method: Principal Axis Factoring.

FACTOR VARIABLES=Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality
 /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE
 /SAVE REG(ALL Operational).

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:37
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=Operational_ BI_Availability Operational_BI_Agility Operational_BI_Quality /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL Operational).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.03
	Maximum Memory Required	2028 (1.980K) bytes
Variables Created	Operational1	Regression factor score 1

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.699
Bartlett's Test of Sphericity	Approx. Chi-Square	141.929
	df	3
	Sig.	.000

Factor Matrix^a

	Factor 1
Operational_BI_Availability	.695
Operational_BI_Agility	.770
Operational_BI_Quality	.731

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 9 iterations required.

Communalities

	Extraction
Operational_BI_Availability	.484
Operational_BI_Agility	.593
Operational_BI_Quality	.534

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.610	53.667	53.667

Extraction Method: Principal Axis Factoring.

FACTOR VARIABLES=BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformBI_Alt
ernative_Solutions

```

/PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTAT
E /SAVE REG(ALL DMQ) .

```

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:37
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=BI_Improves _Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /PRINT=KMO EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL DMQ).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	3096 (3.023K) bytes
Variables Created	DMQ1	Regression factor score 1

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.813
Bartlett's Test of Sphericity	Approx. Chi-Square	278.222
	df	6
	Sig.	.000

Factor Matrix^a

	Factor 1
BI_Improves_Decisions	.745
BI_Speed_Accuracy	.835
BI_Data_Inform	.755
BI_Alternative_Solutions	.693

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 7 iterations required.

Communalities

	Extraction
BI_Improves_Decisions	.555
BI_Speed_Accuracy	.697
BI_Data_Inform	.570
BI_Alternative_Solutions	.480

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	2.303	57.569	57.569

Extraction Method: Principal Axis Factoring.

* ===== STEP 3: Harman single-factor (CMB). =====.

FACTOR VARIABLES=Strategic_BI_AvailabilityStrategic_BI_AlternativesStrategic
 _BI_Alignment
 Tactical_BI_AvailabilityTactical_BI_Resource_AllocTactical_BI_Data_Decisio
 ns
 Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality
 BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformBI_Alternative_Soluti
 ons
 /PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:37
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		FACTOR VARIABLES=Strategic_BI _Availability Strategic_BI_Alternatives Strategic_BI_Alignment Tactical_BI_Availability Tactical_BI_Resource_All oc Tactical_BI_Data_Decisio ns Operational_BI_Availabilit y Operational_BI_Agility Operational_BI_Quality BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	21700 (21.191K) bytes

Factor Matrix^a

	Factor 1
Strategic_BI_Availability	.663
Strategic_BI_Alternatives	.759
Strategic_BI_Alignment	.700
Tactical_BI_Availability	.707
Tactical_BI_Resource_Alloc	.728
Tactical_BI_Data_Decisions	.664
Operational_BI_Availability	.670
Operational_BI_Agility	.694
Operational_BI_Quality	.689
BI_Improves_Decisions	.776
BI_Speed_Accuracy	.738
BI_Data_Inform	.718
BI_Alternative_Solutions	.653

Extraction Method: Principal Axis
Factoring.

a. 1 factors extracted. 4 iterations required.

Communalities

	Extraction
Strategic_BI_Availability	.440
Strategic_BI_Alternatives	.576
Strategic_BI_Alignment	.490
Tactical_BI_Availability	.499
Tactical_BI_Resource_Alloc	.530
Tactical_BI_Data_Decisions	.441
Operational_BI_Availability	.448
Operational_BI_Agility	.482
Operational_BI_Quality	.475
BI_Improves_Decisions	.602
BI_Speed_Accuracy	.544
BI_Data_Inform	.515
BI_Alternative_Solutions	.427

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	6.471	49.776	49.776

Extraction Method: Principal Axis Factoring.

* ===== STEP 4: descriptives+ formal normality test. =====.
 DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality
 /STATISTICS=MEAN STDDEV SKEWNESS KURTOSIS.

Descriptives

Notes

Output Created		18-JUN-2026 01:25:37
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality /STATISTICS=MEAN STDDEV SKEWNESS KURTOSIS.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Descriptive Statistics

	N Statistic	Mean Statistic	Std. Deviation Statistic	Skewness		Kurtosis Statistic
				Statistic	Std. Error	
Strategic_Mean	178	3.7341	.77820	-.759	.182	1.045
Tactical_Mean	178	3.7978	.70798	-.791	.182	2.051
Operational_Mean	178	3.8258	.74679	-.720	.182	.556
Decision_Making_Quality	178	3.8244	.75334	-.659	.182	.380
Valid N (listwise)	178					

Descriptive Statistics

	Kurtosis Std. Error
Strategic_Mean	.362
Tactical_Mean	.362
Operational_Mean	.362
Decision_Making_Quality	.362
Valid N (listwise)	

EXAMINE VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality
/PLOT NPLOT /STATISTICS NONE /NOTOTAL.

Explore

Notes

Output Created		18-JUN-2026 01:25:37
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values for dependent variables are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.

Notes

Syntax	EXAMINE VARIABLES=Strategic_M ean Tactical_Mean Operational_Mean Decision_Making_Quality /PLOT NPLOT /STATISTICS NONE /NOTOTAL.	
Resources	Processor Time	00:00:01.28
	Elapsed Time	00:00:01.79

Case Processing Summary

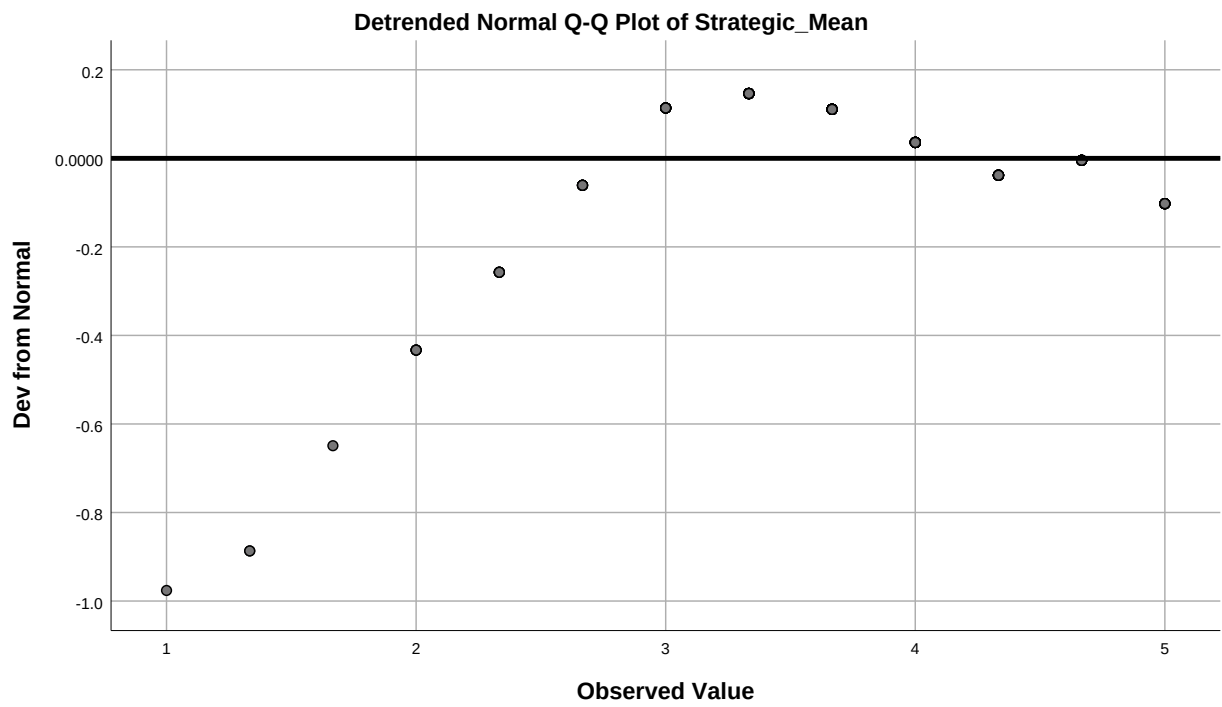
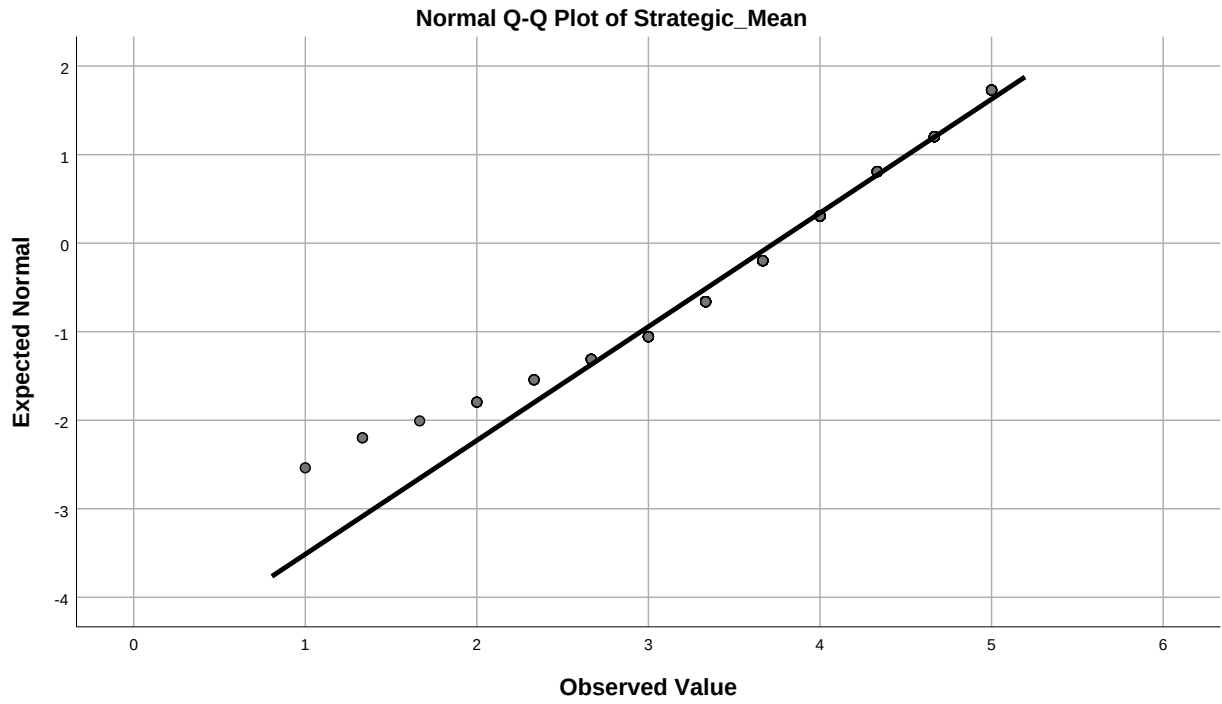
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Strategic_Mean	178	100.0%	0	0.0%	178	100.0%
Tactical_Mean	178	100.0%	0	0.0%	178	100.0%
Operational_Mean	178	100.0%	0	0.0%	178	100.0%
Decision_Making_Quality	178	100.0%	0	0.0%	178	100.0%

Tests of Normality

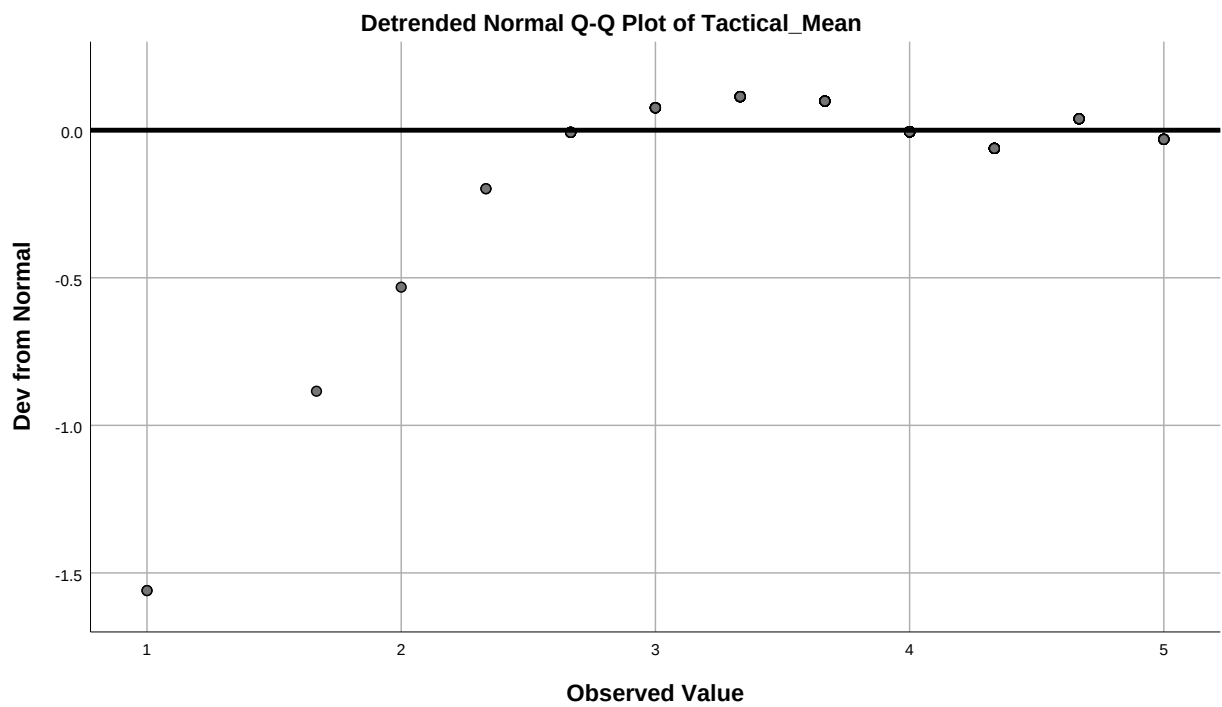
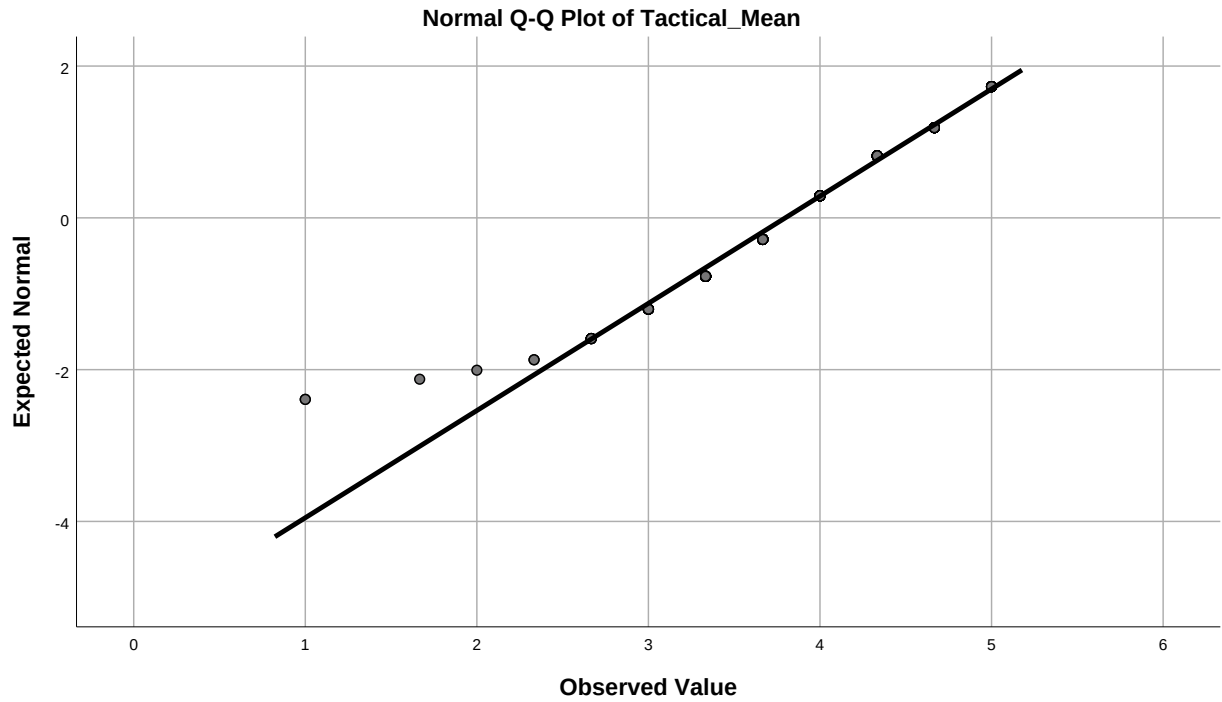
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Strategic_Mean	.134	178	.000	.943	178	.000
Tactical_Mean	.140	178	.000	.935	178	.000
Operational_Mean	.176	178	.000	.944	178	.000
Decision_Making_Quality	.154	178	.000	.953	178	.000

a. Lilliefors Significance Correction

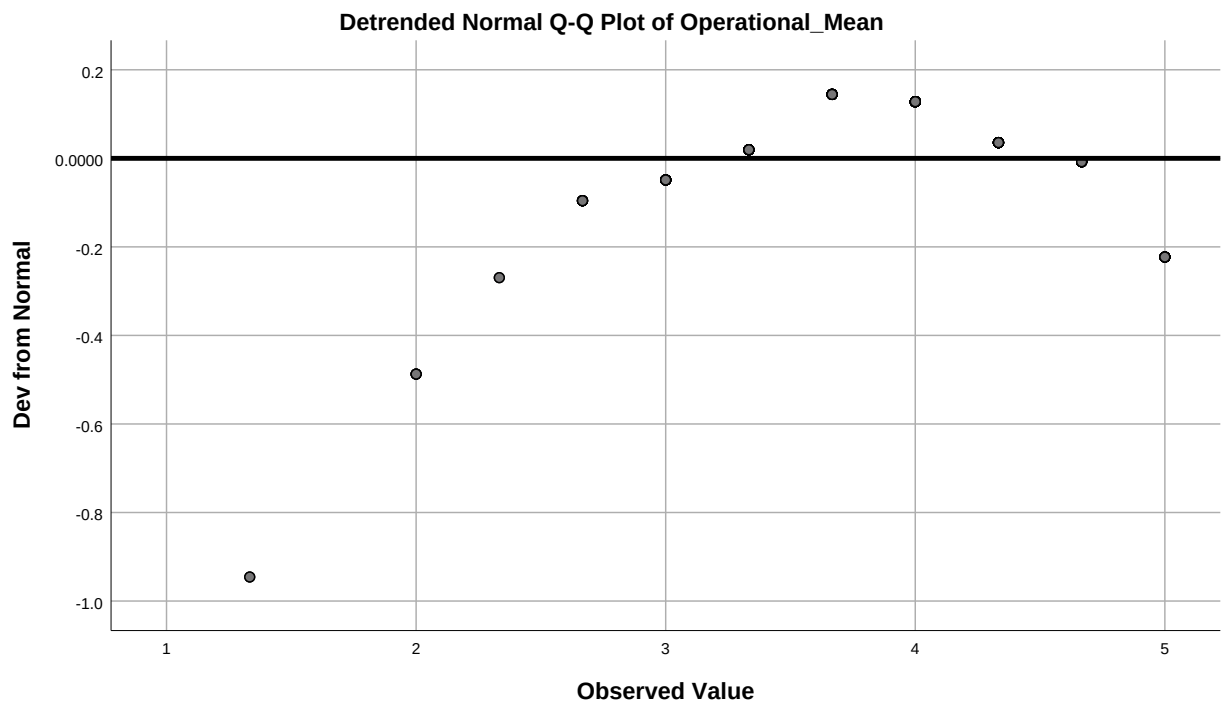
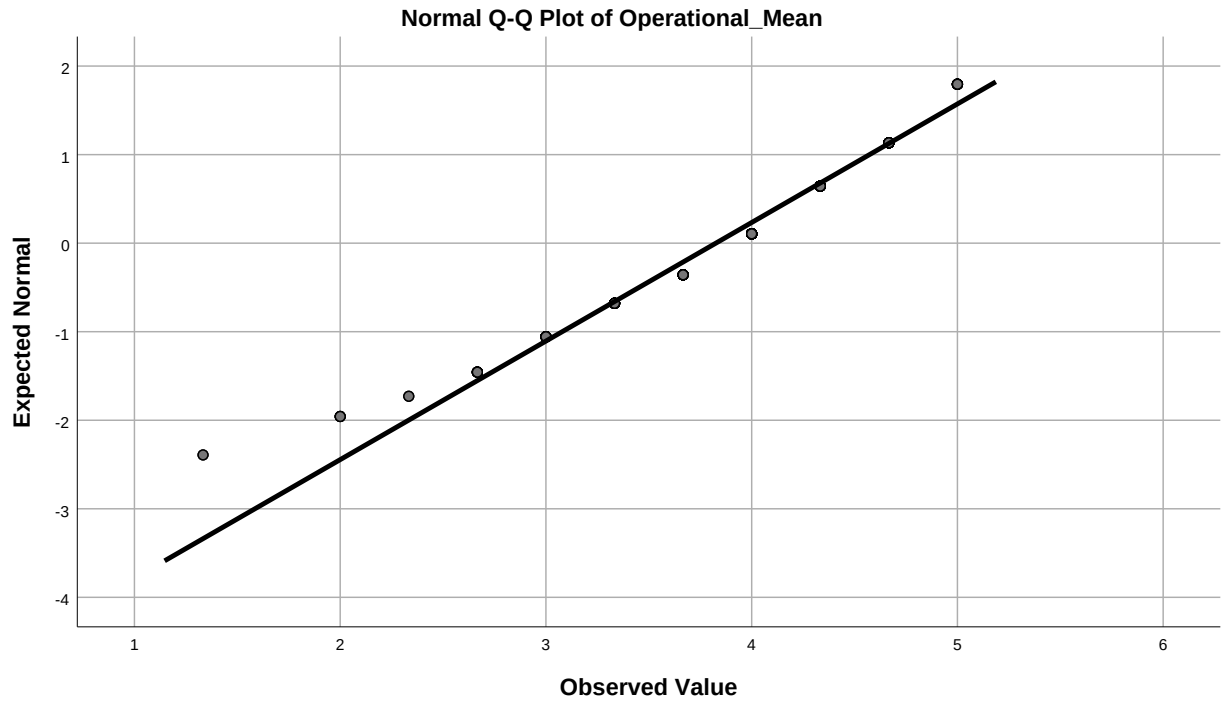
Strategic_Mean



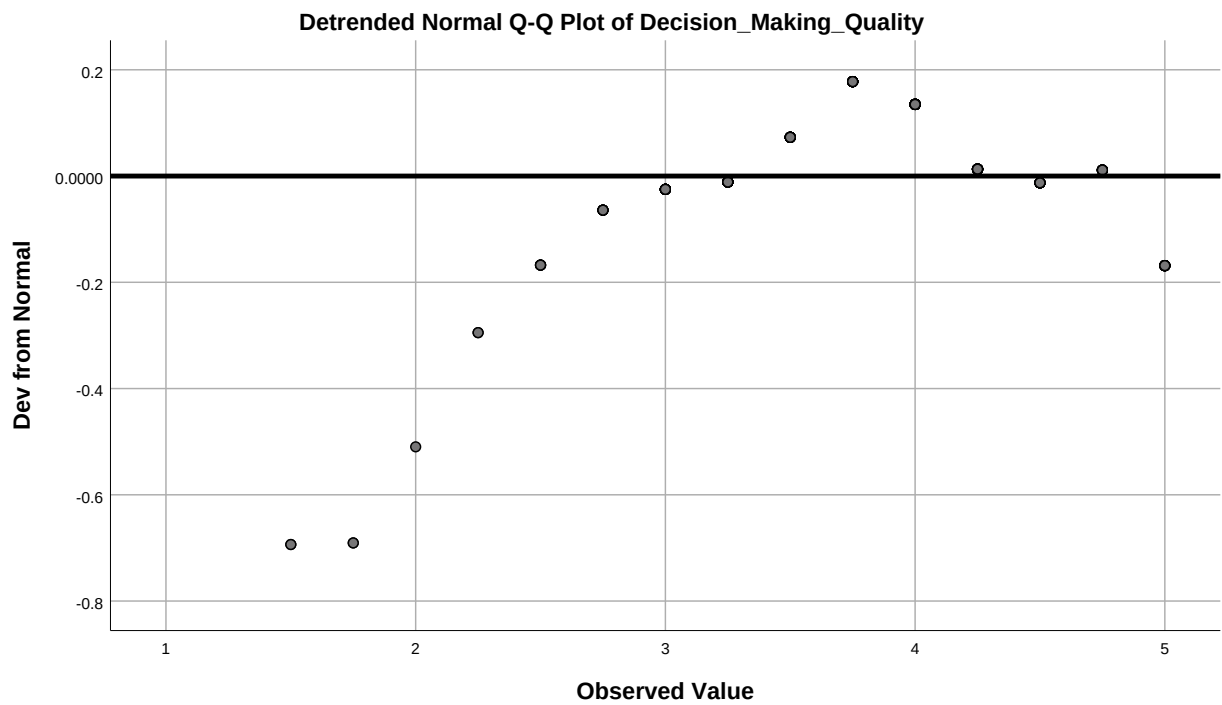
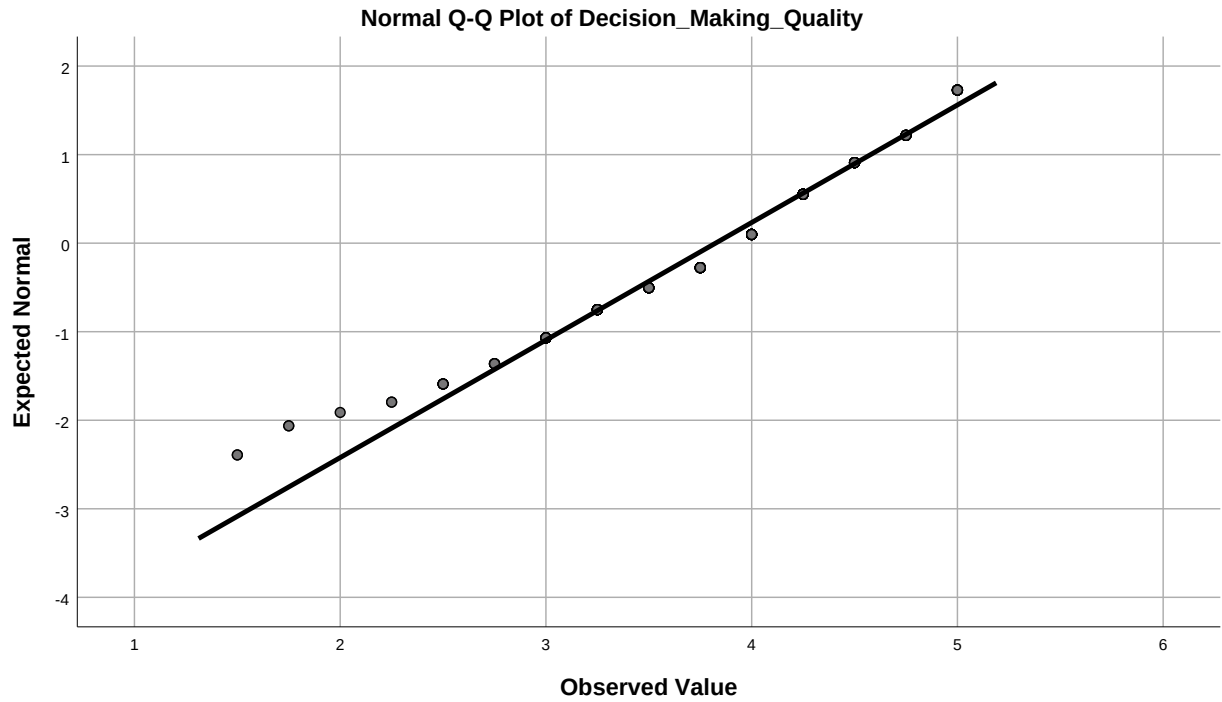
Tactical_Mean



Operational_Mean



Decision_Making_Quality



* ===== STEP 5: Spearman correlations (non-normal data). =====.

NONPAR CORR VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality
 /PRINT=SPEARMAN TWOTAIL SIG.

Nonparametric Correlations

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		NONPAR CORR VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality /PRINT=SPEARMAN TWOTAIL SIG.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Number of Cases Allowed	241979 cases ^a

a. Based on availability of workspace memory

Correlations

			Strategic_Mean	Tactical_Mean
Spearman's rho	Strategic_Mean	Correlation Coefficient	1.000	.681
		Sig. (2-tailed)	.	.000
		N	178	178
	Tactical_Mean	Correlation Coefficient	.681	1.000
		Sig. (2-tailed)	.000	.
		N	178	178
	Operational_Mean	Correlation Coefficient	.583	.641
		Sig. (2-tailed)	.000	.000
		N	178	178
	Decision_Making_Quality	Correlation Coefficient	.630	.623
		Sig. (2-tailed)	.000	.000
		N	178	178

Correlations

			Operational_Mean	Decision_Making_Quality
Spearman's rho	Strategic_Mean	Correlation Coefficient	.583	.630
		Sig. (2-tailed)	.000	.000
		N	178	178
	Tactical_Mean	Correlation Coefficient	.641	.623
		Sig. (2-tailed)	.000	.000
		N	178	178
	Operational_Mean	Correlation Coefficient	1.000	.668
		Sig. (2-tailed)	.	.000
		N	178	178
	Decision_Making_Quality	Correlation Coefficient	.668	1.000
		Sig. (2-tailed)	.000	.
		N	178	178

- * ===== STEP 6: MAIN regression on factor scores + assumptions. =====.
- * Multicollinearity(VIF/TOL), independence (Durbin-Watson), residual.
- * normality (histogram, normal P-P), homoscedasticity/linearity (scatter), .
- * influence (Cook's, Mahalanobis).

REGRESSION

```

/STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL
/DEPENDENT DMQ1
/METHOD=ENTER Strategic1 Tactical1 Operational1
/RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
/SCATTERPLOT=(*ZRESID ,*ZPRED)
/SAVE COOK MAHAL.

```

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS CI(95) R ANOVA COLLIN TOL /DEPENDENT DMQ1 /METHOD=ENTER Strategic1 Tactical1 Operational1 /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /SCATTERPLOT= (*ZRESID ,*ZPRED) /SAVE COOK MAHAL.
Resources	Processor Time	00:00:00.20
	Elapsed Time	00:00:00.32

Notes

	Memory Required	2540 bytes
	Additional Memory Required for Residual Plots	896 bytes
Variables Created or Modified	MAH_1	Mahalanobis Distance
	COO_1	Cook's Distance

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis ^b	.	Enter

a. Dependent Variable: REGR factor score 1 for analysis

b. All requested variables entered.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.791 ^a	.626	.619	.56980776	1.634

a. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

b. Dependent Variable: REGR factor score 1 for analysis

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	94.466	3	31.489	96.984	.000 ^b
	Residual	56.494	174	.325		
	Total	150.961	177			

a. Dependent Variable: REGR factor score 1 for analysis

b. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-6.023E-17	.043		.000	1.000
	REGR factor score 1 for analysis	.263	.073	.258	3.614	.000
	REGR factor score 1 for analysis	.242	.079	.232	3.079	.002
	REGR factor score 1 for analysis	.419	.068	.400	6.124	.000

Coefficients^a

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-.084	.084		
	REGR factor score 1 for analysis	.119	.406	.421	2.377
	REGR factor score 1 for analysis	.087	.397	.380	2.629
	REGR factor score 1 for analysis	.284	.554	.503	1.988

a. Dependent Variable: REGR factor score 1 for analysis

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	(Constant)	Variance Proportions	
					REGR factor score 1 for analysis	REGR factor score 1 for analysis
1	1	2.367	1.000	.00	.06	.06
	2	1.000	1.539	1.00	.00	.00
	3	.378	2.503	.00	.32	.06
	4	.255	3.049	.00	.62	.89

Collinearity Diagnostics^a

Model	Dimension	Variance ... REGR factor score 1 for analysis
1	1	.07
	2	.00
	3	.87
	4	.06

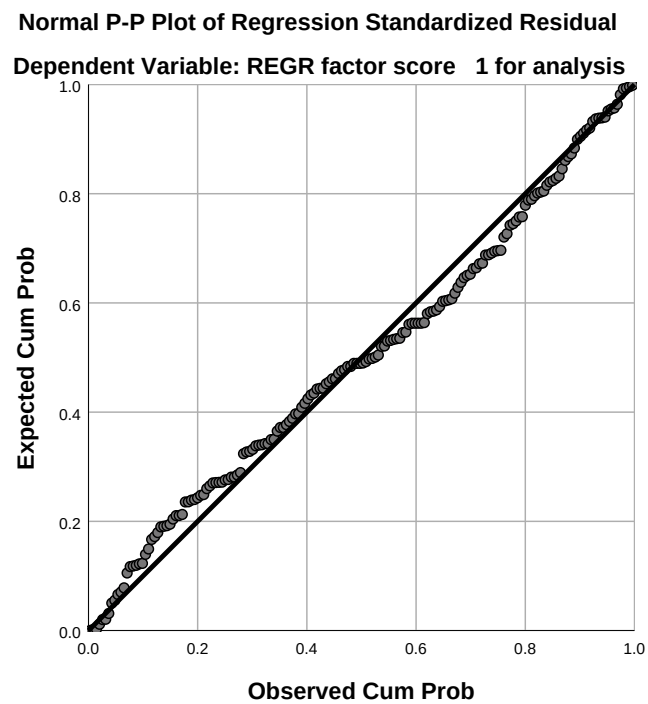
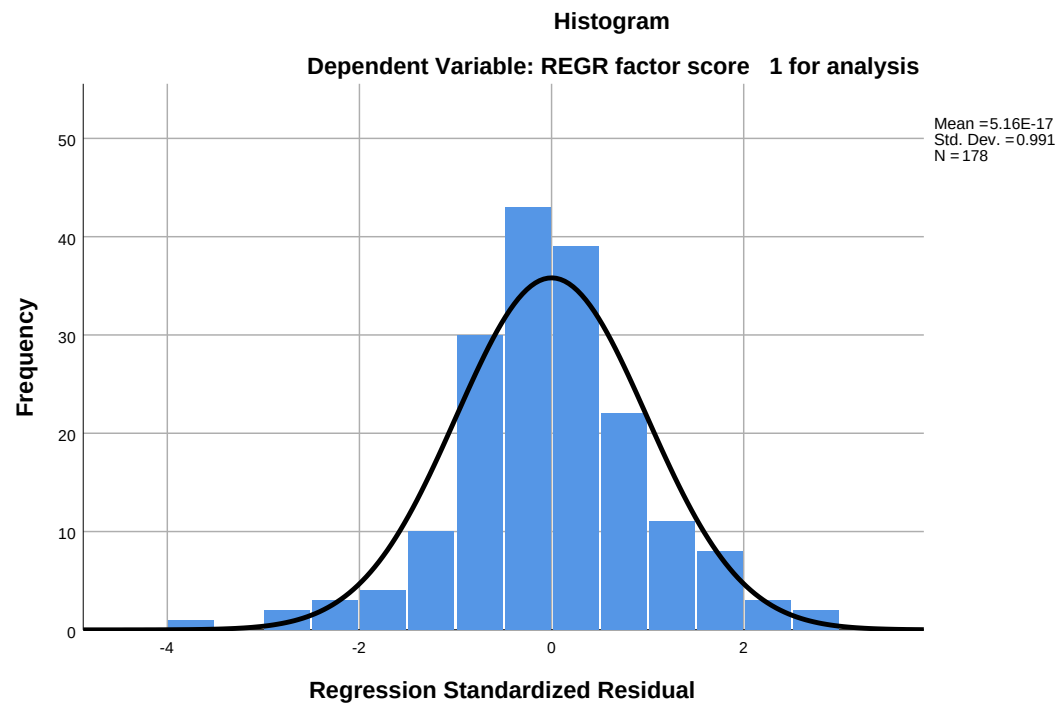
a. Dependent Variable: REGR factor score 1 for analysis

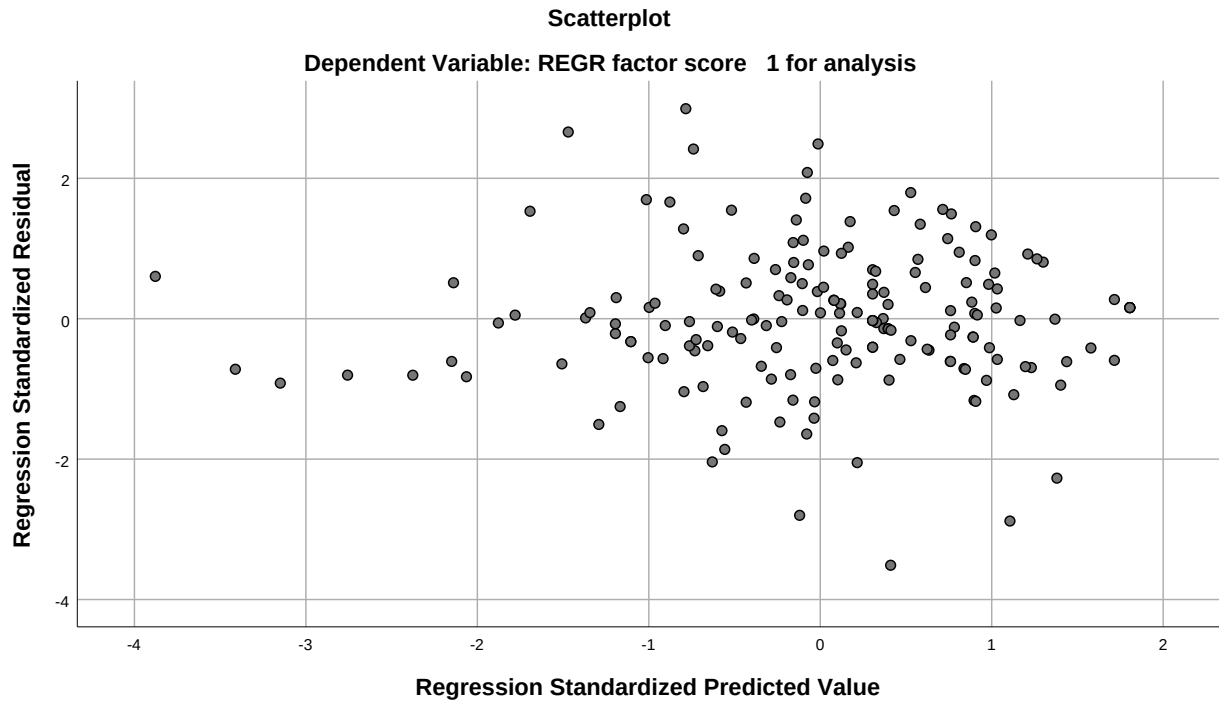
Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	-2.8331993	1.3189460	.0000000	.73055277	178
Std. Predicted Value	-3.878	1.805	.000	1.000	178
Standard Error of Predicted Value	.045	.180	.081	.028	178
Adjusted Predicted Value	-2.8704939	1.3167009	-.0013313	.73017782	178
Residual	-2.00097680	1.70448911	.00000000	.56495824	178
Std. Residual	-3.512	2.991	.000	.991	178
Stud. Residual	-3.533	3.023	.001	1.004	178
Deleted Residual	-2.02583694	1.74111760	.00133135	.57926667	178
Stud. Deleted Residual	-3.657	3.097	.001	1.014	178
Mahal. Distance	.111	16.672	2.983	3.045	178
Cook's Distance	.000	.151	.006	.015	178
Centered Leverage Value	.001	.094	.017	.017	178

a. Dependent Variable: REGR factor score 1 for analysis

Charts





* ===== STEP 7: robustness - composite means. =====.

```

REGRESSION
  /STATISTICS COEFF OUTS R ANOVA
  /DEPENDENT Decision_Making_Quality
  /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean
  
```

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS R ANOVA /DEPENDENT Decision_Making_Quality /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	2572 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Operational_Mean, Strategic_Mean, Tactical_Mean ^b	.	Enter

a. Dependent Variable: Decision_Making_Quality

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.791 ^a	.626	.619	.46483

a. Predictors: (Constant), Operational_Mean, Strategic_Mean, Tactical_Mean

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	62.856	3	20.952	96.970	.000 ^b
	Residual	37.595	174	.216		
	Total	100.451	177			

a. Dependent Variable: Decision_Making_Quality

b. Predictors: (Constant), Operational_Mean, Strategic_Mean, Tactical_Mean

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.419	.204		2.056	.041
	Strategic_Mean	.257	.070	.266	3.672	.000
	Tactical_Mean	.281	.081	.264	3.468	.001
	Operational_Mean	.360	.067	.357	5.335	.000

a. Dependent Variable: Decision_Making_Quality

* ===== STEP 8: BI challenges (Table) + crosstabs. =====.

```
DESCRIPTIVES VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_I
ssue BI_Flexibility_Issue
/STATISTICS=MEAN STDDEV.
```

Descriptives

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.

Notes

Syntax	DESCRIPTIVES VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /STATISTICS=MEAN STDDEV.	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Descriptive Statistics

	N	Mean	Std. Deviation
BI_Training_Limit	178	3.76	.970
BI_Cost_Barrier	178	3.52	1.059
BI_Data_Integration_Issue	178	3.47	.909
BI_Flexibility_Issue	178	3.32	.999
Valid N (listwise)	178		

FREQUENCIES VARIABLES=BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue

Frequencies

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Statistics

		BI_Training_Limit	BI_Cost_Barrier	BI_Data_Integration_Issue	BI_Flexibility_Issue
N	Valid	178	178	178	178
	Missing	0	0	0	0

Frequency Table

BI_Training_Limit

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	2	1.1	1.1	1.1
	2	18	10.1	10.1	11.2
	3	44	24.7	24.7	36.0
	4	71	39.9	39.9	75.8
	5	43	24.2	24.2	100.0
	Total	178	100.0	100.0	

BI_Cost_Barrier

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	8	4.5	4.5	4.5
	2	21	11.8	11.8	16.3
	3	52	29.2	29.2	45.5
	4	65	36.5	36.5	82.0
	5	32	18.0	18.0	100.0
	Total	178	100.0	100.0	

BI_Data_Integration_Issue

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	4	2.2	2.2	2.2
	2	18	10.1	10.1	12.4
	3	67	37.6	37.6	50.0
	4	68	38.2	38.2	88.2
	5	21	11.8	11.8	100.0
	Total	178	100.0	100.0	

BI_Flexibility_Issue

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	7	3.9	3.9	3.9
	2	26	14.6	14.6	18.5
	3	70	39.3	39.3	57.9
	4	53	29.8	29.8	87.6
	5	22	12.4	12.4	100.0
	Total	178	100.0	100.0	

CROSSTABS TABLES=Job_Position BY BI_Training_LimitBI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue/CELLS=ROW.

Crosstabs

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each table are based on all the cases with valid data in the specified range(s) for all variables in each table.

Notes

Syntax	CROSSTABS TABLES=Job_Position BY BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /CELLS=ROW.	
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Dimensions Requested	2
	Cells Available	349496

Case Processing Summary

	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Job_Position * BI_Training_Limit	178	100.0%	0	0.0%	178	100.0%
Job_Position * BI_Cost_Barrier	178	100.0%	0	0.0%	178	100.0%
Job_Position * BI_Data_Integration_Issue	178	100.0%	0	0.0%	178	100.0%
Job_Position * BI_Flexibility_Issue	178	100.0%	0	0.0%	178	100.0%

Job_Position * BI_Training_Limit Crosstabulation

% within Job_Position

		BI_Training_Limit					Total
		1	2	3	4	5	
Job_Position	1		15.4%	30.8%	7.7%	46.2%	100.0%
	2		4.8%	38.1%	38.1%	19.0%	100.0%
	3		5.6%	25.0%	52.8%	16.7%	100.0%
	4		10.0%	26.7%	50.0%	13.3%	100.0%
	5	2.6%	12.8%	19.2%	35.9%	29.5%	100.0%
Total		1.1%	10.1%	24.7%	39.9%	24.2%	100.0%

Job_Position * BI_Cost_Barrier Crosstabulation

% within Job_Position

		BI_Cost_Barrier					Total
		1	2	3	4	5	
Job_Position	1		7.7%	46.2%	23.1%	23.1%	100.0%
	2		14.3%	28.6%	47.6%	9.5%	100.0%
	3	8.3%	8.3%	22.2%	47.2%	13.9%	100.0%
	4		10.0%	30.0%	46.7%	13.3%	100.0%
	5	6.4%	14.1%	29.5%	26.9%	23.1%	100.0%
Total		4.5%	11.8%	29.2%	36.5%	18.0%	100.0%

Job_Position * BI_Data_Integration_Issue Crosstabulation

% within Job_Position

		BI_Data_Integration_Issue					Total
		1	2	3	4	5	
Job_Position	1	7.7%	7.7%	30.8%	38.5%	15.4%	100.0%
	2		9.5%	52.4%	33.3%	4.8%	100.0%
	3	2.8%	11.1%	36.1%	38.9%	11.1%	100.0%
	4		10.0%	30.0%	46.7%	13.3%	100.0%
	5	2.6%	10.3%	38.5%	35.9%	12.8%	100.0%
Total		2.2%	10.1%	37.6%	38.2%	11.8%	100.0%

Job_Position * BI_Flexibility_Issue Crosstabulation

% within Job_Position

		BI_Flexibility_Issue					Total
		1	2	3	4	5	
Job_Position	1	15.4%	23.1%	30.8%	15.4%	15.4%	100.0%
	2	4.8%	9.5%	38.1%	38.1%	9.5%	100.0%
	3	2.8%	16.7%	30.6%	38.9%	11.1%	100.0%
	4		10.0%	50.0%	26.7%	13.3%	100.0%
	5	3.8%	15.4%	41.0%	26.9%	12.8%	100.0%
Total		3.9%	14.6%	39.3%	29.8%	12.4%	100.0%

CROSSTABS TABLES=Industry BY BI_Training_LimitBI_Cost_BarrierBI_Data_Integration_Issue BI_Flexibility_Issue/CELLS=ROW.

Crosstabs

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each table are based on all the cases with valid data in the specified range(s) for all variables in each table.
Syntax		CROSSTABS TABLES=Industry BY BI_Training_Limit BI_Cost_Barrier BI_Data_Integration_Issue BI_Flexibility_Issue /CELLS=ROW.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Dimensions Requested	2
	Cells Available	349496

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Industry * BI_Training_Limit	178	100.0%	0	0.0%	178	100.0%
Industry * BI_Cost_Barrier	178	100.0%	0	0.0%	178	100.0%
Industry * BI_Data_Integration_Issue	178	100.0%	0	0.0%	178	100.0%
Industry * BI_Flexibility_Issue	178	100.0%	0	0.0%	178	100.0%

Industry * BI_Training_Limit Crosstabulation

% within Industry

		BI_Training_Limit					Total
		1	2	3	4	5	
Industry	1		7.7%	15.4%	38.5%	38.5%	100.0%
	2	2.2%	10.9%	32.6%	37.0%	17.4%	100.0%
	3	1.9%	7.7%	25.0%	48.1%	17.3%	100.0%
	4		10.0%	20.0%	36.7%	33.3%	100.0%
	5		13.5%	21.6%	35.1%	29.7%	100.0%
Total		1.1%	10.1%	24.7%	39.9%	24.2%	100.0%

Industry * BI_Cost_Barrier Crosstabulation

% within Industry

		BI_Cost_Barrier					Total
		1	2	3	4	5	
Industry	1	15.4%	7.7%	15.4%	38.5%	23.1%	100.0%
	2		15.2%	41.3%	37.0%	6.5%	100.0%
	3	5.8%	13.5%	25.0%	30.8%	25.0%	100.0%
	4	3.3%	6.7%	16.7%	50.0%	23.3%	100.0%
	5	5.4%	10.8%	35.1%	32.4%	16.2%	100.0%
Total		4.5%	11.8%	29.2%	36.5%	18.0%	100.0%

Industry * BI_Data_Integration_Issue Crosstabulation

% within Industry

		BI_Data_Integration_Issue					Total
		1	2	3	4	5	
Industry	1	7.7%	7.7%	7.7%	53.8%	23.1%	100.0%
	2	2.2%	10.9%	37.0%	37.0%	13.0%	100.0%
	3	1.9%	11.5%	44.2%	30.8%	11.5%	100.0%
	4		3.3%	33.3%	50.0%	13.3%	100.0%
	5	2.7%	13.5%	43.2%	35.1%	5.4%	100.0%
Total		2.2%	10.1%	37.6%	38.2%	11.8%	100.0%

Industry * BI_Flexibility_Issue Crosstabulation

% within Industry

		BI_Flexibility_Issue					Total
		1	2	3	4	5	
Industry	1		15.4%	38.5%	30.8%	15.4%	100.0%
	2	6.5%	13.0%	47.8%	23.9%	8.7%	100.0%
	3	1.9%	19.2%	42.3%	19.2%	17.3%	100.0%
	4	3.3%	10.0%	20.0%	53.3%	13.3%	100.0%
	5	5.4%	13.5%	40.5%	32.4%	8.1%	100.0%
Total		3.9%	14.6%	39.3%	29.8%	12.4%	100.0%

* ===== STEP 9: sensitivity - drop the most BI-referential DMQ item. ==.
 FACTOR VARIABLES=BI_Speed_AccuracyBI_Data_InformBI_Alternative_Solutions
 /MISSING LISTWISE /PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL DMQ3).

Factor Analysis

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.
Syntax		FACTOR VARIABLES=BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /MISSING LISTWISE /PRINT=EXTRACTION /CRITERIA=FACTORS(1) /EXTRACTION=PAF /ROTATION=NOROTATE /SAVE REG(ALL DMQ3).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Maximum Memory Required	2028 (1.980K) bytes
Variables Created	DMQ31	Regression factor score 1

Factor Matrix^a

	Factor 1
BI_Speed_Accuracy	.811
BI_Data_Inform	.764
BI_Alternative_Solutions	.708

Extraction Method: Principal Axis
Factoring.

a. 1 factors extracted. 9 iterations required.

Communalities

	Extraction
BI_Speed_Accuracy	.658
BI_Data_Inform	.583
BI_Alternative_Solutions	.501

Extraction Method: Principal Axis
Factoring.

Total Variance Explained

Factor	Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	1.743	58.103	58.103

Extraction Method: Principal Axis Factoring.

REGRESSION

/STATISTICS COEFF OUTS CI(95) R ANOVA

/DEPENDENT DMQ31

/METHOD=ENTER Strategic1 Tactical1 Operational1

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS CI(95) R ANOVA /DEPENDENT DMQ31 /METHOD=ENTER Strategic1 Tactical1 Operational1.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	2588 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis ^b	.	Enter

a. Dependent Variable: REGR factor score 1 for analysis

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.751 ^a	.564	.556	.60051463

a. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	81.044	3	27.015	74.913	.000 ^b
	Residual	62.748	174	.361		
	Total	143.792	177			

a. Dependent Variable: REGR factor score 1 for analysis

b. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-4.115E-17	.045		.000	1.000
	REGR factor score 1 for analysis	.230	.077	.232	3.009	.003
	REGR factor score 1 for analysis	.237	.083	.233	2.864	.005
	REGR factor score 1 for analysis	.388	.072	.380	5.379	.000

Coefficients^a

Model		95.0% Confidence Interval for B	
		Lower Bound	Upper Bound
1	(Constant)	-.089	.089
	REGR factor score 1 for analysis	.079	.382
	REGR factor score 1 for analysis	.074	.401
	REGR factor score 1 for analysis	.246	.530

a. Dependent Variable: REGR factor score 1 for analysis

```

* ===== STEP 10: robustness - demographic controls (incl. size). =====.
RECODE Job_Position (1=1)(ELSE=0) INTO Pos_CEO.
RECODE Job_Position (2=1)(ELSE=0) INTO Pos_Senior.
RECODE Job_Position (3=1)(ELSE=0) INTO Pos_Middle.
RECODE Job_Position (4=1)(ELSE=0) INTO Pos_Operational.
RECODE Industry (1=1)(ELSE=0) INTO Ind_Manuf.
RECODE Industry (2=1)(ELSE=0) INTO Ind_Retail.
RECODE Industry (3=1)(ELSE=0) INTO Ind_Service.
RECODE Industry (4=1)(ELSE=0) INTO Ind_Tech.
RECODE Company_Size (2=1)(ELSE=0) INTO Size_50_99.
RECODE Company_Size (3=1)(ELSE=0) INTO Size_99_250.
EXECUTE.
REGRESSION

```

```

/STATISTICS COEFF OUTS R ANOVA COLLIN TOL
/DEPENDENT DMQ1
/METHOD=ENTER Strategic1 Tactical1 Operational1
  Pos_CEO Pos_Senior Pos_Middle Pos_Operational
  Ind_Manuf Ind_Retail Ind_Service Ind_Tech
  Size_50_99 Size_99_250.

```

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS R ANOVA COLLIN TOL /DEPENDENT DMQ1 /METHOD=ENTER Strategic1 Tactical1 Operational1 Pos_CEO Pos_Senior Pos_Middle Pos_Operational Ind_Manuf Ind_Retail Ind_Service Ind_Tech Size_50_99 Size_99_250.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Notes

Memory Required	7468 bytes
Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Size_99_250, Ind_Service, REGR factor score 1 for analysis, Pos_Senior, Pos_CEO, Ind_Manuf, Pos_Middle, Ind_Tech, Pos_Operational, Size_50_99, REGR factor score 1 for analysis, Ind_Retail, REGR factor score 1 for analysis ^b	.	Enter

a. Dependent Variable: REGR factor score 1 for analysis

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.805 ^a	.647	.619	.56978257

a. Predictors: (Constant), Size_99_250, Ind_Service, REGR factor score 1 for analysis, Pos_Senior, Pos_CEO, Ind_Manuf, Pos_Middle, Ind_Tech, Pos_Operational, Size_50_99, REGR factor score 1 for analysis, Ind_Retail, REGR factor score 1 for analysis

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	97.718	13	7.517	23.153	.000 ^b
	Residual	53.243	164	.325		
	Total	150.961	177			

a. Dependent Variable: REGR factor score 1 for analysis

b. Predictors: (Constant), Size_99_250, Ind_Service, REGR factor score 1 for analysis, Pos_Senior, Pos_CEO, Ind_Manuf, Pos_Middle, Ind_Tech, Pos_Operational, Size_50_99, REGR factor score 1 for analysis, Ind_Retail, REGR factor score 1 for analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.060	.121		.492	.623
	REGR factor score 1 for analysis	.274	.075	.270	3.672	.000
	REGR factor score 1 for analysis	.259	.081	.247	3.202	.002
	REGR factor score 1 for analysis	.389	.073	.372	5.309	.000
	Pos_CEO	.025	.189	.007	.134	.894
	Pos_Senior	.173	.147	.060	1.171	.243
	Pos_Middle	-.108	.125	-.047	-.858	.392
	Pos_Operational	-.028	.139	-.011	-.202	.841
	Ind_Manuf	.276	.199	.078	1.389	.167
	Ind_Retail	-.120	.130	-.057	-.924	.357
	Ind_Service	.044	.126	.022	.347	.729
	Ind_Tech	-.038	.151	-.016	-.254	.800
	Size_50_99	-.007	.112	-.004	-.066	.948
	Size_99_250	-.169	.115	-.083	-1.460	.146

Coefficients^a

Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	REGR factor score 1 for analysis	.398	2.511
	REGR factor score 1 for analysis	.361	2.773
	REGR factor score 1 for analysis	.437	2.286
	Pos_CEO	.751	1.332
	Pos_Senior	.807	1.239
	Pos_Middle	.719	1.390
	Pos_Operational	.673	1.485
	Ind_Manuf	.683	1.465
	Ind_Retail	.561	1.781
	Ind_Service	.553	1.808
	Ind_Tech	.570	1.755
	Size_50_99	.635	1.574
	Size_99_250	.669	1.494

a. Dependent Variable: REGR factor score 1 for analysis

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	(Constant)	Variance Proportions	
					REGR factor score 1 for analysis	REGR factor score 1 for analysis
1	1	3.324	1.000	.01	.00	.00
	2	2.477	1.159	.00	.04	.04
	3	1.356	1.566	.00	.00	.00
	4	1.207	1.660	.00	.00	.00
	5	1.037	1.791	.00	.01	.01
	6	.982	1.840	.00	.00	.00
	7	.895	1.928	.00	.00	.00
	8	.834	1.997	.00	.02	.00
	9	.698	2.182	.00	.00	.01
	10	.357	3.052	.00	.27	.04
	11	.317	3.240	.00	.01	.02
	12	.238	3.737	.00	.62	.86
	13	.196	4.121	.02	.02	.01
	14	.084	6.290	.96	.01	.01

Collinearity Diagnostics^a

Model	Dimension	REGR factor score 1 for analysis	Variance Proportions			
			Pos_CEO	Pos_Senior	Pos_Middle	Pos_Operational
1	1	.00	.00	.01	.01	.01
	2	.05	.01	.00	.00	.00
	3	.00	.04	.00	.06	.07
	4	.00	.21	.01	.00	.02
	5	.00	.02	.09	.09	.00
	6	.00	.03	.43	.04	.06
	7	.00	.00	.01	.03	.00
	8	.00	.06	.01	.06	.25
	9	.01	.27	.03	.11	.01
	10	.59	.00	.01	.01	.01
	11	.16	.04	.16	.30	.09
	12	.04	.02	.01	.01	.00
	13	.11	.29	.21	.25	.45
	14	.03	.02	.01	.02	.02

Collinearity Diagnostics^a

Model	Dimension	Variance Proportions					
		Ind_Manuf	Ind_Retail	Ind_Service	Ind_Tech	Size_50_99	Size_99_250
1	1	.01	.01	.01	.01	.02	.01
	2	.00	.01	.00	.00	.00	.00
	3	.02	.01	.04	.05	.02	.05
	4	.19	.01	.03	.01	.01	.01
	5	.01	.12	.04	.01	.03	.06
	6	.00	.04	.01	.01	.01	.01
	7	.11	.04	.01	.15	.04	.16
	8	.18	.06	.01	.00	.01	.00
	9	.02	.01	.15	.12	.01	.00
	10	.02	.00	.04	.04	.09	.09
	11	.03	.06	.00	.04	.53	.08
	12	.00	.00	.01	.01	.01	.02
	13	.25	.19	.22	.32	.02	.26
	14	.15	.45	.43	.23	.21	.24

a. Dependent Variable: REGR factor score 1 for analysis

```
* ===== STEP 11: scope robustness - firms <=99 employees. =====.
TEMPORARY.
SELECT IF Company_Size <= 2.
REGRESSION
  /STATISTICS COEFF OUTS R ANOVA
  /DEPENDENT DMQ1
  /METHOD=ENTER Strategic1 Tactical1 Operational1
```

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	127
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS R ANOVA /DEPENDENT DMQ1 /METHOD=ENTER Strategic1 Tactical1 Operational1.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00
	Memory Required	2788 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis ^b	.	Enter

a. Dependent Variable: REGR factor score 1 for analysis

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.820 ^a	.673	.665	.53753543

a. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	73.148	3	24.383	84.385	.000 ^b
	Residual	35.540	123	.289		
	Total	108.688	126			

a. Dependent Variable: REGR factor score 1 for analysis

b. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.038	.048		.794	.429
	REGR factor score 1 for analysis	.284	.079	.277	3.589	.000
	REGR factor score 1 for analysis	.158	.087	.148	1.815	.072
	REGR factor score 1 for analysis	.512	.078	.490	6.552	.000

a. Dependent Variable: REGR factor score 1 for analysis

```

* ===== STEP 12: response-style sensitivity. =====.
* Excludes the 8 retained partial straight-liners (identical answer to.
* all 13 BI/DMQ items but varied on the challenge items).
COMPUTE bi_sd = SD(Strategic_BI_AvailabilityStrategic_BI_AlternativesStrateg
ic_BI_Alignment,
Tactical_BI_AvailabilityTactical_BI_Resource_AllocationTactical_BI_Data_Decisio
ns,
Operational_BI_AvailabilityOperational_BI_AgilityOperational_BI_Quality
BI_Improves_DecisionsBI_Speed_AccuracyBI_Data_InformationBI_Alternative_Soluti
ons).
EXECUTE.
TEMPORARY.
SELECT IF bi_sd > 0.
REGRESSION
  /STATISTICS COEFF OUTS R ANOVA
  /DEPENDENT DMQ1
  /METHOD=ENTER Strategic1 Tactical1 Operational1

```

Regression

Notes

Output Created		18-JUN-2026 01:25:39
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	170
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /STATISTICS COEFF OUTS R ANOVA /DEPENDENT DMQ1 /METHOD=ENTER Strategic1 Tactical1 Operational1.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	2812 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis ^b	.	Enter

a. Dependent Variable: REGR factor score 1 for analysis

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.776 ^a	.602	.595	.58310863

a. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	85.296	3	28.432	83.620	.000 ^b
	Residual	56.443	166	.340		
	Total	141.739	169			

a. Dependent Variable: REGR factor score 1 for analysis

b. Predictors: (Constant), REGR factor score 1 for analysis, REGR factor score 1 for analysis, REGR factor score 1 for analysis

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-.001	.045		-.030	.976
	REGR factor score 1 for analysis	.261	.074	.257	3.508	.001
	REGR factor score 1 for analysis	.240	.081	.227	2.964	.003
	REGR factor score 1 for analysis	.417	.070	.397	5.932	.000

a. Dependent Variable: REGR factor score 1 for analysis

* ===== STEP 13: group comparisons (optional, non-parametric). =====.

NPAR TESTS /K-W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Company_Size(1 3).

NPar Tests

Notes

Output Created		18-JUN-2026 01:25:40
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Syntax		NPAR TESTS /K- W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Company_Size(1 3).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00
	Number of Cases Allowed ^a	157286

a. Based on availability of workspace memory.

Kruskal-Wallis Test

Ranks

	Company_Size	N	Mean Rank
Strategic_Mean	1	65	75.08
	2	62	104.33
	3	51	89.84
	Total	178	
Tactical_Mean	1	65	74.48
	2	62	101.15
	3	51	94.47
	Total	178	
Operational_Mean	1	65	68.48
	2	62	107.52
	3	51	94.39
	Total	178	
Decision_Making_Quality	1	65	76.84
	2	62	104.54
	3	51	87.35
	Total	178	

Test Statistics^{a,b}

	Strategic_Mean	Tactical_Mean	Operational_Mean	Decision_Making_Quality
Kruskal-Wallis H	10.468	9.443	19.343	9.435
df	2	2	2	2
Asymp. Sig.	.005	.009	.000	.009

a. Kruskal Wallis Test

b. Grouping Variable: Company_Size

NPAR TESTS /K-W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Job_Position(1 5).

NPar Tests

Notes

Output Created		18-JUN-2026 01:25:40
Comments		
Input	Data	C: \Users\Hoda\OneDrive\De sktop\GIT\BI_DMQ_data_ coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Syntax		NPAR TESTS /K- W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Job_Position(1 5).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00
	Number of Cases Allowed ^a	157286

a. Based on availability of workspace memory.

Kruskal-Wallis Test

Ranks

	Job_Position	N	Mean Rank
Strategic_Mean	1	13	87.62
	2	21	86.10
	3	36	98.29
	4	30	95.67
	5	78	84.30
	Total	178	
Tactical_Mean	1	13	62.42
	2	21	83.74
	3	36	99.54
	4	30	103.30
	5	78	85.62
	Total	178	
Operational_Mean	1	13	89.77
	2	21	91.95
	3	36	99.08
	4	30	105.43
	5	78	78.24
	Total	178	
Decision_Making_Quality	1	13	80.54
	2	21	102.81
	3	36	95.69
	4	30	97.47
	5	78	81.49
	Total	178	

Test Statistics^{a,b}

	Strategic_Mean	Tactical_Mean	Operational_Mean	Decision_Making_Quality
Kruskal-Wallis H	2.437	8.050	8.086	4.991
df	4	4	4	4
Asymp. Sig.	.656	.090	.088	.288

a. Kruskal Wallis Test

b. Grouping Variable: Job_Position

NPAR TESTS /K-W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Industry(1 5).

NPar Tests

Notes		
Output Created		18-JUN-2026 01:25:40
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\GIT\BI_DMQ_data_coded.sav
	Active Dataset	DataSet178
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	178
Syntax		NPAR TESTS /K-W=Strategic_Mean Tactical_Mean Operational_Mean Decision_Making_Quality BY Industry(1 5).
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.00
	Number of Cases Allowed ^a	157286

a. Based on availability of workspace memory.

Kruskal-Wallis Test

Ranks

	Industry	N	Mean Rank
Strategic_Mean	1	13	98.58
	2	46	85.22
	3	52	80.99
	4	30	114.32
	5	37	83.47
	Total	178	
Tactical_Mean	1	13	94.96
	2	46	80.12
	3	52	88.83
	4	30	102.95
	5	37	89.28
	Total	178	
Operational_Mean	1	13	108.08
	2	46	67.66
	3	52	91.41
	4	30	114.27
	5	37	87.35
	Total	178	
Decision_Making_Quality	1	13	109.81
	2	46	72.26
	3	52	90.44
	4	30	106.90
	5	37	88.36
	Total	178	

Test Statistics^{a,b}

	Strategic_Mean	Tactical_Mean	Operational_Mean	Decision_Making_Quality
Kruskal-Wallis H	9.832	3.837	17.455	10.783
df	4	4	4	4
Asymp. Sig.	.043	.428	.002	.029

a. Kruskal Wallis Test

b. Grouping Variable: Industry

```
* =====.
* NOTE - validity analyses NOT done here (run BI_DMQ_validity.py):.
* convergent (AVE, CR), discriminant (HTMT, Fornell-Larcker), and the.
* four-factor vs one-factor CFA. Base SPSS cannot estimate CFA.
* =====.
* End of syntax (n=178).
```